Milestone1

Team Name: Team Cool Guys

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Database Requirement Specification

Our team is trying to design a database for a meal swipe donation system since there are a lot of students who cannot finish their swipes at the end of each semester, and students who need more food as well. Our database is designed to keep the information of the users of the donation system and keep track of all the transactions.

We intend to include **Users**, **Listings**, **Transactions**, **Status**, **Location**, **School**, **Type**, **Urgency**, **Discount**, and **Comment**. **Users** will have First Name, Last Name, Phone Number, and Time of Creation attributes, uniquely identified by their NetID. Also a foreign key to the school entity: School ID. A **Listing** will refer to a User as a seller, and will have a posted date, price, amount of swipes to sell, expiration date, and a boolean telling us if it is active. It will refer to a table of locations by PreferredLocationID and will have a unique ListingID. It also has foreign keys: TypeID, UrgencyID, and DiscountID. Each attempted **transactions** will be stored within the transaction table, which will reference a listing, a buyer, and a status. It is uniquely

identified by the transactionID. Each transaction will also have a transaction_time, buyer_confirmation, and seller_confirmation fields. Each transaction will be linked to a **Status** table, which includes a status_name and will be uniquely identified by a statusID. Each **Location** will be identified by a locationID, and has a location name field. Each **School** instance will have a primary key School_ID, as well as a field called SchoolName. Each **Type** instance will have a TypeID as a primary key, a Type attribute tells what the type of listing actually is, also a SemesterValid field. An **Urgency** instance will have a primary key called UrgencyID, and an UrgencyType that tells how urgent the listing is. An **Discount** instance will have a DiscountID primary key, a DiscountRate field, a BeginDate, and a EndDate field. The **Comment** entity will have a CommentID as a primary key, and attributes like TransactionID as a foreign to the transaction table, a Rating field for the transaction, and a CommentText for recording the comment.

Each user can start multiple transactions, although not every transaction will be completed. But each transaction can only be started by one user. Each user can also post multiple listings, and each listing can have at most one preferred location. One listing only belongs to one user. Each Transaction has exactly one Status; each Status can apply to many Transactions.

There can be zero preferred location in the listing, and the buyer and seller can figure out it later. Each Listing must have exactly one Seller (User).

Each comment can only belong to one transaction, and one transaction can only one comment record for all the comments and rating.

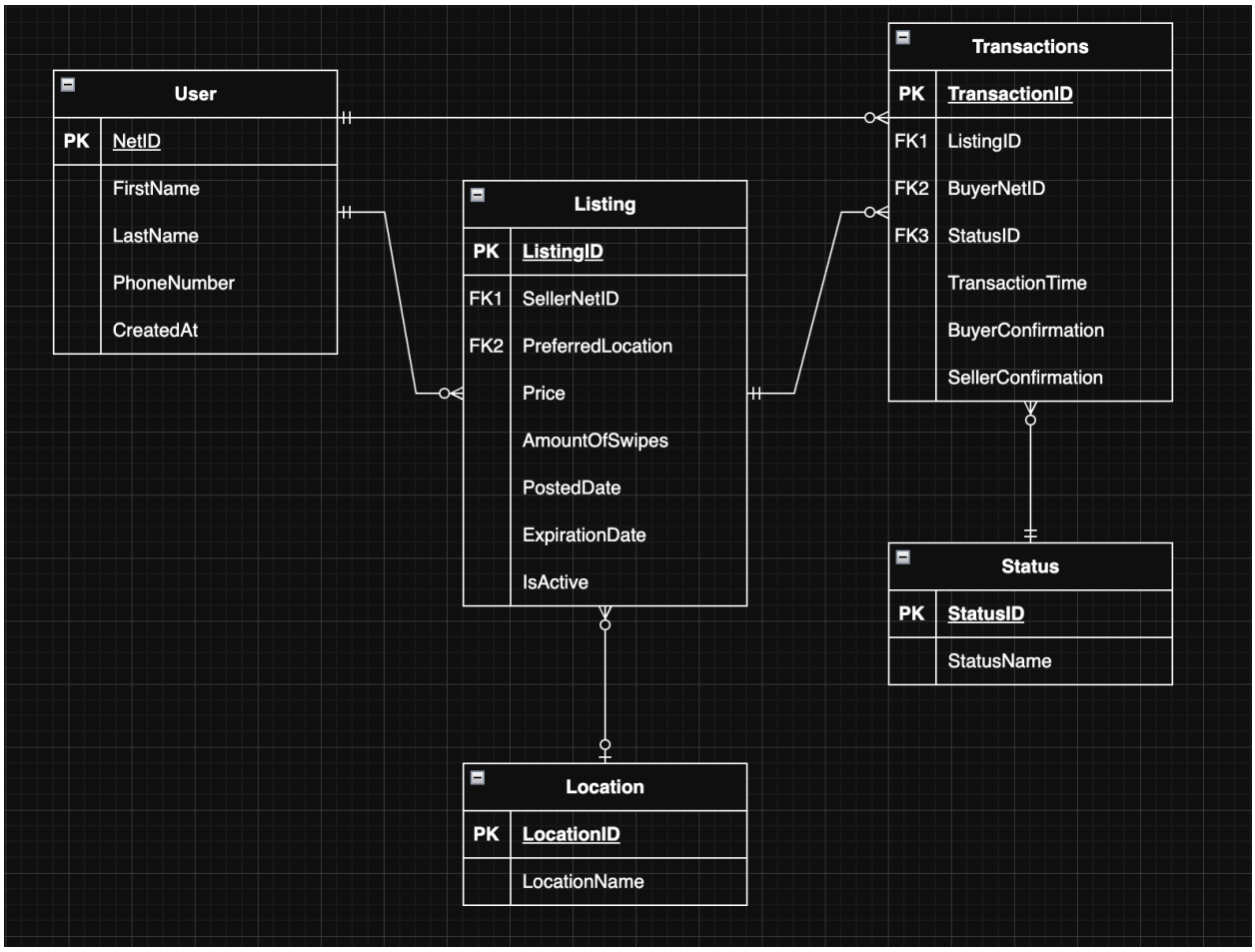
Each Type can be corresponding to zero or many Listing, but each listing can only have one and only one type.

Each Urgency record can only belong to one and only one listing record, and each listing can have one and only one urgency record.

Each discount record can only belong to one and only one listing record, and each listing must have one and only one discount record.

Each School can have zero or many Users, while each user can only belong to one and only one school.

Entity Relationship Diagram(Old)



https://drive.google.com/file/d/1i4dNBQry4nOhoateI3j7QJI61WFwXmG-/view?usp=sharing

Schema Statements(OLD)

User

(NetID, FirstName, LastName, PhoneNumber, CreatedAt)

Listing

(ListingID, @SellerNetID, @PreferredLocation, Price, AmountOfSwipes, PostedDate, ExpirationDate, IsActive)

Transactions

(TransactionID, @ListingID, @BuyerNetID, @StatusID, TransactionTime, BuyerConfirmation, SellerConfirmation)

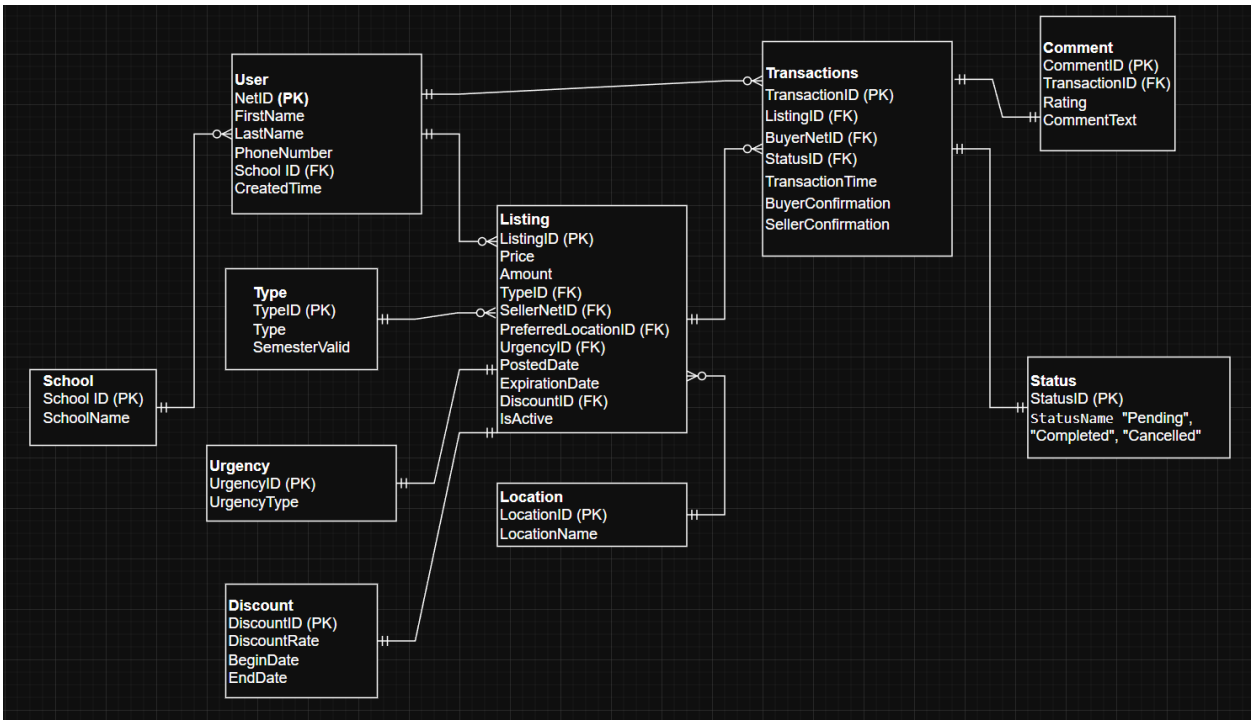
Location

(LocationID, LocationName)

Status

(StatusID, StatusName)

FINAL ERD and Schema Statements for Resubmission (New)



After reconsidering the necessary requirements for our project, we've decided to include more entities into our database. The new ERD is described below, and the new descriptions is updated.

Schema Statements (New)

User

(NetID, FirstName, LastName, PhoneNumber, @School ID, CreatedTime)

School

(School ID, SchoolName)

Listing

(ListingID, @TypeID, @SellerNetID, @PreferredLocation, @UrgencyID, @DiscountID, Price, Amount, PostedDate, ExpirationDate, IsActive)

Transactions

(TransactionID, @ListingID, @BuyerNetID, @StatusID, TransactionTime, BuyerConfirmation, SellerConfirmation)

Location

(LocationID, LocationName)

Status

(StatusID, StatusName)

Urgency

(UrgencyID, UrgencyType)

Discount

(DiscountID, DiscountRate, BeginDate, EndDate)

Comment

(CommentID, @TransactionID, Rating, CommentText)

Type

(TypeID, Type, SemesterValue)

Assumptions

1. All transactions will be recorded regardless of their state.
2. A listing can exist without a transaction, one transaction, or multiple transactions. Only one transaction can be ‘completed’ for each listing in which the buyer buys the entire listing.
3. Each listing can only choose at most one preferred location.
4. Both the buyer and seller are instances belonging to the entity User, although they are represented with different name in listing and transaction tables.

Documentation (Not part of the final milestone, just for documentation)

Felix:

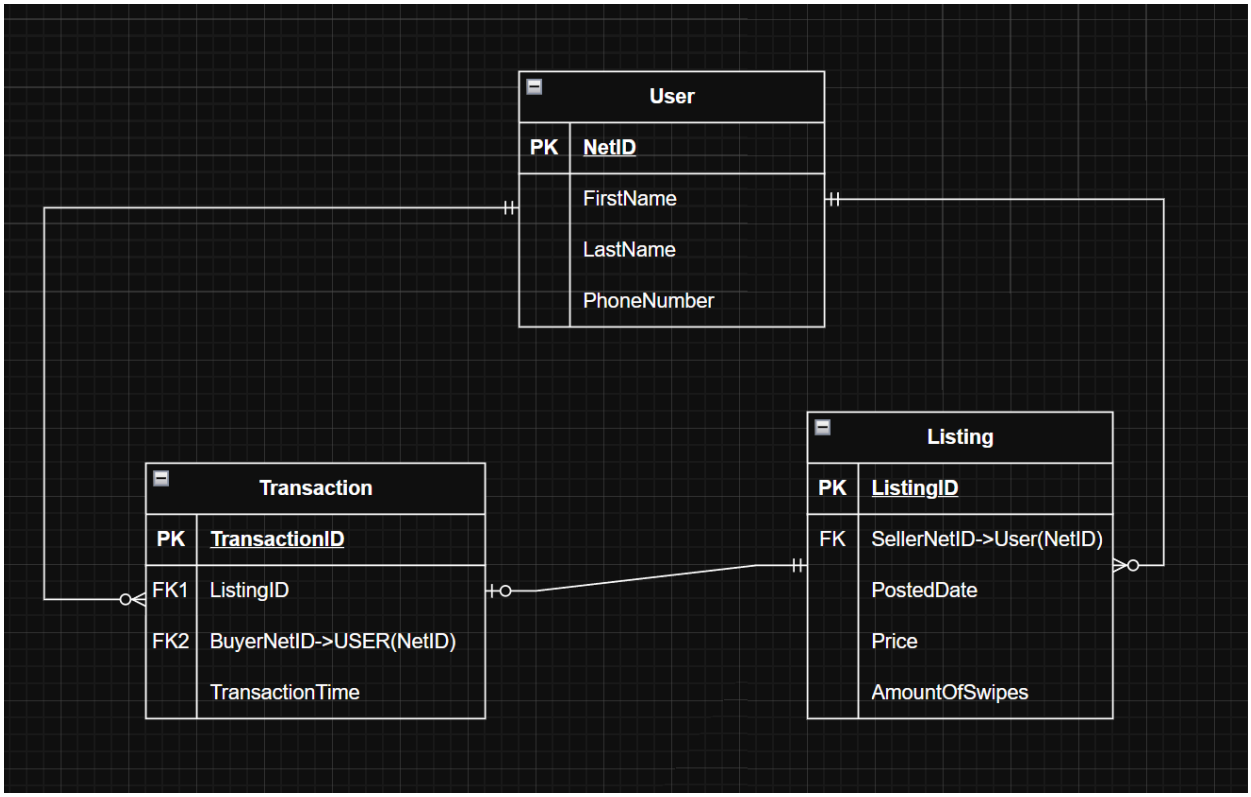
I added some additional features in listing and transactions in my version. In listings, I added preferred location, expiry date, and isActive; in transactions, I added status and buyer/ seller confirmation

Victor:

I kept the ERD minimalistic. also, i think listing to transaction should be 1 to 1 (?) cuz imo ur not supposed to buy multiple listing from one transaction

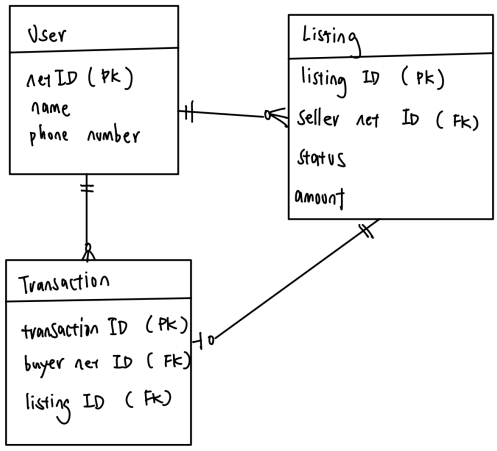
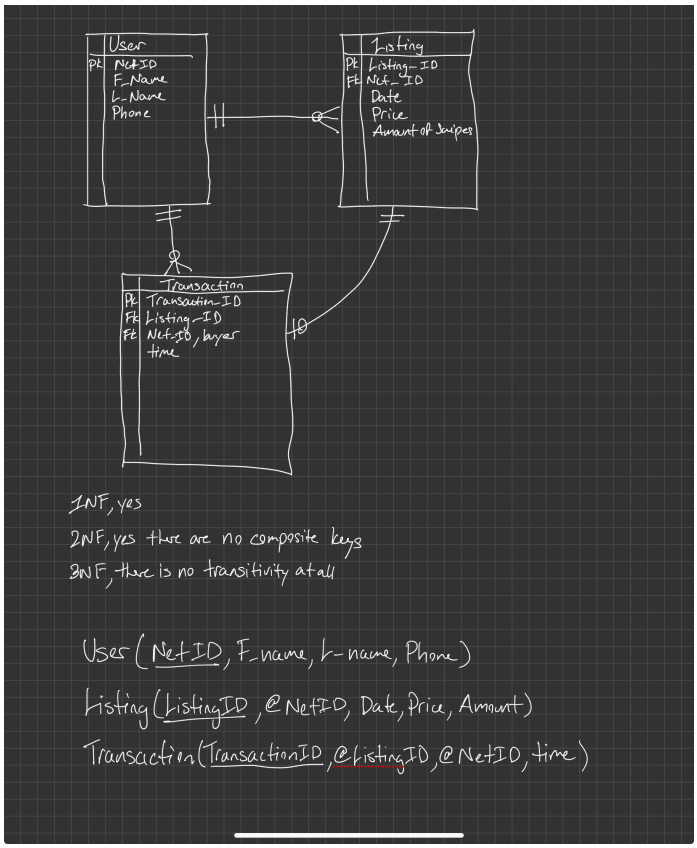
Figo:

I made the ERD according to the schema we created, and checked it is actually already 3NF.



Victor

PROJECT



Alessandro